Strategy Land

Richard Rumelt's Web Journal on Good and Bad Strategy in Business, Politics, and Economics

---

How Bonuses Promote Excessive Risk-Taking

Head I win tails you lose compensation boosts risk-taking

September 6, 2009 By Richard Rumelt

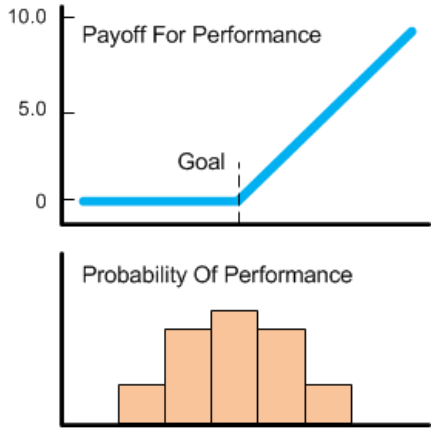
Franklin Delano Raines

In 2002 Fannie Mae gave CEO Franklin Raines a salary of just under $1 million, a bonus of $3.3 million, stock options worth $6.7 million, and a "long-term" incentive payment of $7.2 million. These non-salary payments were each geared to meeting various annual goals for earnings and earnings per share. This very typical way of compensating top executives dramatically increases the riskiness of the strategies they choose.

One way to increase your income under such a plan is to cheat. That is what Clinton-appointee and Rhodes Scholar Franklin Raines did. In 2004, the SEC determined that his administration had overstated the corporation's income by at least $10 billion. He resigned, keeping his past bonuses and retiring on a pension of $114,000 per month.

We all sense that leaving Raines with most of his ill-gotten gains was a mistake. Not only was it unfair to those who didn't cheat, it gave a clear signal to others that cheating pays. But there is more here than a morality play. Raines worked under a standard bonus arrangement that encourages unreasonable risk-taking. One risk he took was of getting caught cooking the books. But he also took huge risks with the public's money by investing heavily in sub-prime mortgages.

Almost all modern senior executive compensation arrangements are like this, creating a focus on the upside coupled with a disdain for the downside. These incentive arrangements actually *encourage* unwise risk-taking. To see why, look at the chart below.



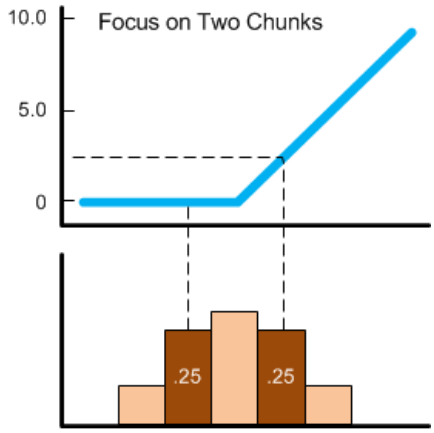
Typical Bonus Plan

In this incentive arrangement the payoff is tied to a *performance measure*. If the performance measure comes in below the *goal*, there is no payoff. Payoffs only happen if the *goal* is met or beaten. And, as the performance measure rises above the *goal*, the payoff rises accordingly.

The key is that the payoff for poor performance is the same as that for disastrous performance. The effects of worse and worse failure are muted. If you are a finance whiz, you will recognize the shape of the payoff curve as being a *call option*. In a call option, the *goal* is called the *strike-price*. If you have purchased a call option, you are then in a position to profit if the underlying stock price rises above the strike price, but your losses are limited no matter how poorly the stock does. Similarly, the executive bonus plan pays nothing for performance less than the goal, but pays out a fraction of each increment of performance above the goal. In other words, it caps downside losses, but allows continuing upside gains.

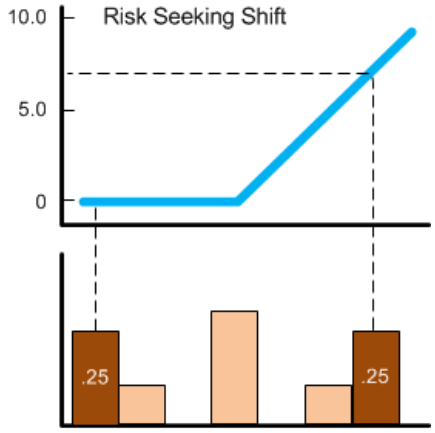
Learning to price options like this is a staple of modern MBA training. Nevertheless, I frequently find that my students, having mastered option pricing formulas in their finance courses, have little intuition about the transformation that options induce. In particular, when you own a call option, especially one where the strike-price is above the current price, or where the goal is difficult and chancy, then you are converted from being a normal risk-avoiding human into an uncertainty-loving mutant.

The basic mechanism is easy to see. The chart below shows the bonus payoff with two chunks of probability highlighted. The left-side chunk represents a 25 percent chance of below-goal performance, yielding zero, and the right-side chunk represents a 25 percent chance of above goal performance, yielding a payout of $2.5 million. Together, these two chunks of probability create an expected payoff of $1.25 million (0.25 x 0 + 0.25 x 2.5).



Isolate Two Chunks of Probability

Now, suppose that the executive can change the probabilities. He might, for example, change from selling ordinary insurance policies to those with payouts tied to the level of the stock market several years hence. This change means more uncertainty in outcomes and, in the chart below, I have modeled this as a very simple movement of the two chunks of probability. The left-side 25% chunk has moved far to the left and the right-side chunk has moved far to the right[1]



Increased Risk Equals More Bonus

With this shift, the left-side chunk of probability still yields a payoff of zero—bad performance always yields zero. But the right-side chunk now yields a payoff of $7 million. The two chunks together create an expected payoff of $3.5 million. By taking more risk, with no increase in expected performance, the executive has dramatically ballooned his expected payoff.

This is a very general phenomenon. Whenever he payoff curve bends upwards (is convex), then an increase in uncertainty increases the expected payoff.[2] Thus, if you own a call option, an increase in uncertainty should make its value rise. The difference between a speculator's call option and a senior executive's bonus arrangement is that the executive is in a position to manipulate the amount of uncertainty. Thus, the bonus arrangement induces a transformation into a risk-taking mutant.

To see an example, consider Bear Stearns' bonus compensation plan. Its top twelve executives received bonuses based on meeting and exceeding certain targeted levels of annual return-on-equity.[3] As the company's market value declined and is leverage soared in late 2007, one solution was to bring in new equity investors. Such a move would dilute existing shareholders, but might just save the over-levered company from complete failure. One reason this did not happen, reports William Cohan[4] was the target-based return-on-equity bonus system. If lots of new equity were added, the denominator of the return-on-equity formula would rise, the targets would be missed, and there would be no bonuses for the senior executives. Also, when one good slug of your annual "bonus" is bigger than the lifetime earnings of the average citizen, maybe the "longer-term" doesn't matter very much.

The standard rationale for these compensation arrangement is "aligning executive interests with those of shareholders." There is a large and flourishing compensation-consulting industry that will happily make up pay and bonus arrangements that are "competitive" with the pay and bonus arrangements they made up last year and those they made for other firms. This rationale of "alignment" is a lie because such arrangements do no such thing.

Upward curving incentive arrangements are the solution to the problem of making people work hard. If executives were loading coal into hoppers, then increasing pay per ton-loaded would help counter the pain of working harder. But executives are not loading coal into hoppers. They are framing problems, judging risks, and deciding which initiatives to undertake and which to forego. The upward curving bonus contract does, perhaps, make an executive work harder at the job, but it also *unavoidably* encourages wasteful risk-taking.

What is the fix? First, it is necessary to be honest—executive and shareholder interests cannot be perfectly aligned. No matter what neat mathematical model is created by researchers eager to garner income from the compensation industry, it cannot be done. One problem is that stock prices are very noisy measure of performance. Waiting for the "long-term" only increases the noise. Another problem is that it is totally unclear what the phrase "shareholder" means. When does a "shareholder" buy and when do they sell? A third is that executives know more than shareholders about the company and its prospects, so the timing of their buying and selling of shares, and of their grants and exercise of options, are advantaged. Most importantly, shareholders are passive investors or speculators whereas executives are in a position to influence the range of outcomes.

Recognizing the impossibility of a complete solutions, a step in the right direction would be sufficient openness and sufficiently incisive outside evaluation that an executive cannot create the illusion of a better future while actually taking on greater risks or allowing critical resources to decay. This also means a board that takes on the responsibility for actually judging the quality of an executive's work rather than applying a mechanical rule based on metrics. If the organization is too complex for the board to make such judgments, then it should be broken into smaller pieces. Finally, it would be wise to limit the upside in bonus arrangements.

Recently, the financial crisis of 2007-08 has starkly illuminated the absurdity of designing these kinds of incentive plan. They encourage wasteful risk-taking and we have seen the impacts of such dysfunctional arrangements in the behaviors of leading banks, mortgage lenders, investment banks, the government sponsored mortgage entities, and, especially, in the bond and security-rating firms. None of this pattern of foolish incentives and gross overpayments would matter that much if the costs of the ensuing failures were all privately borne. However, these flawed incentive arrangements have created problems that spill across institutions and markets, dragging down the lives and wealth of people with only distant connections to the organizations in question.

Footnotes

1. Technically, this is called a mean-preserving increase in risk. The "mean" being preserved by this symmetrical spreading of probability is of performance, not the payoff. [↩]
2. This symmetrical increase in uncertainty is called *a mean-preserving* spread. [↩]
3. Bear Stearns Companies Inc., 8-K for 2/14/2008. [↩]
4. *House of Cards*, Doubleday, 2009, p. 398. [↩]

Tagged incentives, management, risk | Add a comment

Strategy Land ©2011 by Richard Rumelt. All rights reserved. Wordpress theme SL2010 by General Imagination